

Two Streams, Four Architectures: A Leakage-Free, Statistically Rigorous Benchmark for Pedestrian Crossing-Intention Prediction on PIE

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Abstract

Anticipating whether a pedestrian is about to cross is central to the safety of advanced driver-assistance and automated-driving systems, and the Pedestrian Intention Estimation (PIE) dataset has become the standard benchmark for the task. We show that the windowing protocol in common use on PIE frequently admits the crossing event into the observation window, so that for much of the positive class the benchmark measures detection rather than prediction. Auditing the protocol against PIE's own per-frame annotations quantifies the problem, and we therefore re-extract observation windows anchored at PIE's crossing-point event with a strictly pre-onset horizon, obtaining a protocol with no verified leakage and, as a side effect, several times the data. On that protocol we evaluate a deliberately parsimonious model consuming only a pedestrian bounding box and the ego-vehicle's speed, subjecting it to bootstrap and pedestrian-cluster confidence intervals, leave-one-set-out cross-validation, multi-seed ablations, a documented hyperparameter search, and a detector-in-the-loop test. We then train four temporal architectures, an LSTM, a Transformer, a gated recurrent unit, and an un-gated recurrent network, through one identical engine with matched search budgets, in order to separate the contribution of the input representation from that of the encoder. The results identify which of the two design choices actually carries this task.

Keywords: pedestrian crossing intention; intention prediction; PIE dataset; temporal leakage; LSTM; Transformer; gated recurrent unit; ego-vehicle speed; ADAS; model comparison

1. Introduction

The World Health Organization estimates 1.19 million road traffic deaths in 2021 and reports that road traffic injury remains the leading cause of death for people aged 5–29 years [1]. Pedestrians carry a share of that burden out of proportion to the protection afforded them, accounting for 23% of road traffic fatalities globally, second only to occupants of four-wheeled vehicles (Figure 1a). In the United States, 7,314 pedestrians were killed in 2023, roughly 49% above the 4,910 recorded in 2014, and the pedestrian share of all traffic fatalities climbed from 15% to 18% over that decade (Figure 1b) [2]. Vehicles have become markedly safer for the people travelling inside them without becoming correspondingly safer for those in front of them.

How this risk distributes matters as much as its size. In 2023, 84% of United States pedestrian fatalities occurred in urban areas, 74% away from intersections, and 77% in

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the dark [2]: the typical fatal encounter is not a signalled crosswalk at midday but an unsignalled crossing in city traffic and poor light. Advanced driver-assistance systems (ADAS) must therefore anticipate a crossing *before* it begins, since braking distance at urban speeds consumes most of the interval between recognition and impact.

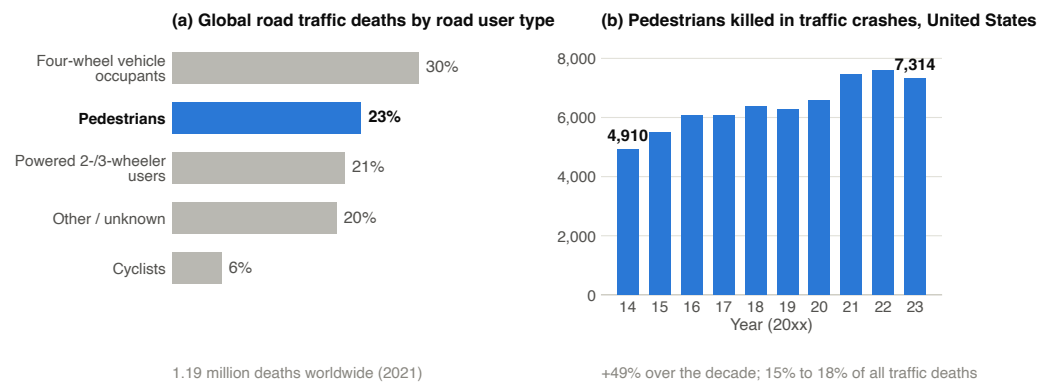


Figure 1. Pedestrians carry a disproportionate and growing share of road traffic casualties. (a) Global road traffic deaths by road user type, 2021 [1]. (b) Pedestrians killed in the United States, 2014–2023 [2].

This is the task of pedestrian crossing-intention prediction: from a short observation of a pedestrian and of the ego-vehicle, estimate the probability that the pedestrian is about to cross [3,4]. It is not trajectory forecasting, which extrapolates motion that has already begun; an intention model must commit to a binary, safety-critical decision while the pedestrian is still on the kerb, under substantial class imbalance. The Pedestrian Intention Estimation (PIE) dataset [3] pairs on-board video with per-frame bounding boxes, per-pedestrian crossing labels, and synchronized on-board-diagnostics (OBD) ego-vehicle speed, and most subsequent models are compared on its fixed split by recording set.

Despite the maturity of this benchmark, three methodological weaknesses recur. First, **temporal leakage is rarely checked**. Many PIE protocols anchor the observation window at the last annotated frame minus a fixed time-to-event offset, with no guarantee that the window ends before the pedestrian begins to cross; when it overlaps frames in which the pedestrian is already crossing, the task quietly collapses from predicting an intention to detecting an event in progress. The dataset authors' own benchmark notes a related misalignment between how intention and action samples are anchored [5], but a direct, quantified audit is, to our knowledge, absent. Second, **recent models are heavy**: three to seven input streams require pose estimators, optical-flow networks, segmentation, or depth models at inference time, which raises cost and is awkward on embedded automotive hardware [6,7]. Third, **evaluation is often thin**, resting on single splits, single-seed ablations reported as conclusions, point estimates without intervals, ground-truth boxes substituted for a real detector, and unjustified hyperparameters [5,8].

We do not claim that minimal-modality prediction is itself new; Section 2 reviews the prior work establishing it. What has been missing is a demonstration that it survives a leakage-free protocol and full statistical scrutiny, together with an explanation of *why* so few inputs suffice.

Our contributions are:

- A quantified **temporal-leakage audit** of the common PIE protocol and a **leakage-free re-extraction** anchored at PIE's own crossing point, verified at 0% window leakage (Sections 3.3 and 5.1).
- A **rigorous re-evaluation** of a parsimonious two-stream (bounding box + ego-speed) model: bootstrap and pedestrian-cluster intervals, leave-one-set-out cross-validation,

multi-seed ablations, a documented hyperparameter search, and a detector-in-the-loop test (Section 5).

- **A four-family architecture study.** An LSTM, a Transformer, a GRU, and an un-gated RNN, trained under one identical engine and compared both at a matched configuration and at equal search budgets, prove statistically indistinguishable on the primary metric, so **the input signal, not the architecture or its gating, decides** (Section 5.3).

Throughout we adopt the metric hierarchy **F1** → **accuracy** → **AUC**: F1 is the imbalance-appropriate operating-point metric we report first, accuracy second, and ROC/PR-AUC as threshold-free corroboration.

2. Related Work

2.1. Datasets, Benchmarks, and Models

Research on crossing prediction has been shaped by a small set of datasets recorded from a moving vehicle. JAAD introduced short clips annotated with pedestrian actions around the moment of crossing [9], and PIE extended this to roughly six hours of continuous urban driving with bounding-box tracks, behavioural tags, and synchronized OBD signals [3]. The benchmark of Kotseruba et al. [4] fixed how observation windows, horizons, and partitions are defined on both and released the PCPA reference model. Even so, results do not transfer cleanly: predictors trained on one dataset lose substantial accuracy on another, pointing to dataset-specific bias rather than a solved task [10]. Two surveys map the area in more detail [5,11].

Early predictors borrowed sequence models from trajectory forecasting, where the LSTM [12] and its bidirectional form [13] had proved effective for pedestrian motion. The field then developed along three lines: graph-based models representing pedestrian, scene, and relations as nodes and edges, from Pedestrian Graph+ [14] to models that treat a group of pedestrians jointly [15]; attention- and transformer-based models that augmented or replaced recurrence, including CAPformer [16], IntFormer [17], PIT [18], PedFormer [19], and GTransPDM [7]; and progressive modality fusion, from PCPA's visual, motion, and ego cues [4] to PIP-Net's seven concurrent streams including depth and optical flow [6]. Recent work adds uncertainty-aware and diffusion formulations [20,21], contextual feature fusion [22], neuro-symbolic reasoning for explainability [23], and multi-context or sequential-plus-visual transformers [24–26]. Sensing beyond the on-board camera has also been tried, including infrastructure-side prediction at intersections [27].

2.2. Minimal-Modality Prediction and the Perception Front End

The prevailing trend adds modalities to chase accuracy, but a counter-current asks whether they earn their cost. The transformer backbone behind most recent models is expensive to run [28], and a crossing predictor must fit inside a tight frame budget on embedded hardware; lighter recurrent units such as the GRU [29] were partly motivated by that pressure. Several works establish that a minimal input is surprisingly strong. Achaji et al. [30] report 91% accuracy and 0.83 F1 on PIE from box geometry alone, though as Section 3.3 shows those numbers were measured under the pre-leakage protocol. Closest to our design, PedCMT predicts crossing from *only* the bounding box and ego-vehicle speed within a cross-modal transformer, on a par with models using more inputs [20], and the pose-free ablation of GTransPDM reaches comparable accuracy from the same two streams [7]. We take minimal inputs as an established design choice and ask how far two cheap streams go once the evaluation is made leakage-free and statistically strict, and *why* they suffice.

Most studies evaluate on ground-truth boxes, which skips the perception stage any deployed system must include. In practice a predictor receives boxes from a single-stage

detector of the YOLO family [31] and identities from a tracking-by-detection method such as ByteTrack [32]. We pair the two to measure how a real front end affects prediction (Section 5.8).

2.3. Pedestrian–Vehicle Interaction and Remaining Gaps

Crossing is an interaction, not a solitary act. A substantial body of work studies how automated vehicles signal intent through external human–machine interfaces and how those signals change crossing decisions [33,34]. That literature concerns *explicit* communication from vehicle to pedestrian; our result points the other way and complements it. The ego-vehicle’s speed profile carries much of the predictive signal, which makes the vehicle’s own motion an *implicit* channel a model can read, while also cautioning that a model leaning on ego-speed partly reads the driver’s anticipation rather than the pedestrian alone.

A fourth gap joins the three set out in the Introduction: the modalities that matter are seldom isolated. A context-aware feature-importance analysis finds that bounding box and ego-speed dominate, and cautions that reliance on ego-speed can induce a driver-side bias in yielding scenarios [8]. Together these four gaps organize the rest of the paper.

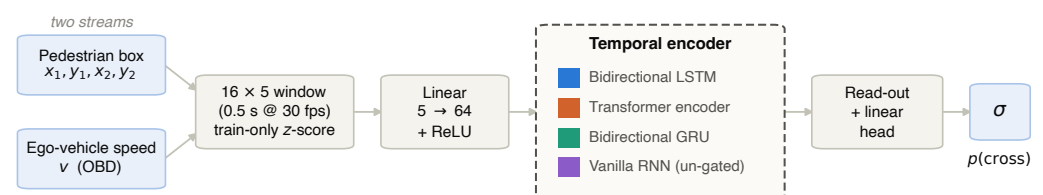
3. Materials and Methods

3.1. Overview

The study has a deliberately narrow design. One dataset, one windowing protocol, one input representation, and one training engine are held fixed; only the factors under test vary, namely the window anchoring rule, the presence of the ego-speed stream, the temporal encoder, the observation length, and the prediction horizon. Everything outside the dashed block of Figure 2a is frozen, which is what allows a difference between two rows of a results table to be attributed to a single design choice.

(a) Two-stream crossing-intention model

Only the dashed encoder block changes across the four families; every other component and the whole training recipe are frozen.



(b) Live perception-to-prediction pipeline

Used for the latency and detector-in-the-loop experiments; the benchmark results use annotated boxes, as is standard.

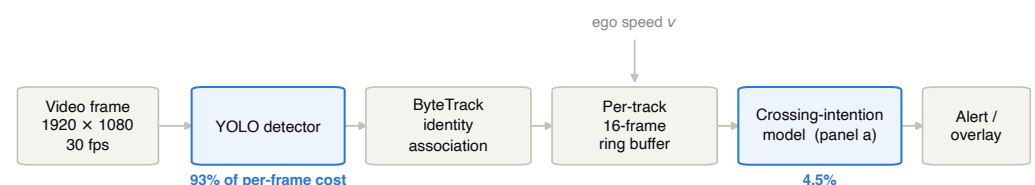


Figure 2. The model and the pipeline it runs inside. (a) The two-stream crossing-intention model; only the dashed encoder block changes across the four families. (b) The live perception-to-prediction pipeline, annotated with each stage’s measured share of per-frame cost.

3.2. The PIE Dataset and Splits

PIE [3] comprises roughly six hours of on-board video (1920×1080 at 30 fps) recorded in Toronto traffic across six recording sets. Each annotated pedestrian carries a per-frame bounding-box track, a per-frame cross state, a binary crossing label and, critically for this work, a `crossing_point` event frame marking when the crossing begins. Ego-vehicle speed is recorded synchronously from the OBD port, giving a second modality available at inference time without additional perception. We use the standard benchmark split by recording set (train set01/02/04, validate set05/06, test set03), which prevents identity- and scene-level leakage because a pedestrian, a video, and a stretch of road each belong to exactly one split, and which is the partition used by the reference benchmark [4] and by our baselines.

3.3. Temporal Leakage and the Leakage-Free Protocol

The prevailing extraction procedure places each observation window at the last annotated frame of a pedestrian's track, minus a fixed time-to-event (TTE) offset. The difficulty is that the end of an annotated track has no defined relationship to the moment the pedestrian steps into the road: a track may continue for many seconds after the crossing begins, in which case a window offset backwards from its end still lands well inside the crossing (Figure 3a), and the task silently changes from predicting an intention to recognizing an event in progress. PIE's per-frame cross attribute lets this be measured directly; we reparsed the annotation XML to recover that state and used it as ground truth for the audit in Section 5.1.

The fix follows from the diagnosis: measure backwards from the event rather than from the end of a track. For each pedestrian we take the contiguous track segment containing `crossing_point`, truncate it at that frame inclusive, and slide 16-frame windows across the remaining history at 50% overlap, keeping only those whose last observed frame falls a TTE of $[30, 60]$ frames (1–2 s) before the crossing point (Figure 3b). Non-crossing pedestrians contribute windows from their full track under the same length and stride, and pedestrians whose usable track is shorter than the window plus the minimum look-ahead (107 of 1,374, or 7.8%) are excluded.

Two properties make this leakage-free by construction rather than by inspection: `crossing_point` coincides with the first frame labelled `cross = crossing` for 516 of 519 crossing pedestrians (99.4%) and never precedes it, and requiring 30 frames of look-ahead puts the last observed frame at least a full second before that marker. Table 1 summarizes what changes; the leakage-free protocol also yields 3.5 times the data, because it slides windows across pre-crossing history rather than taking one per track.

Table 1. Track-end versus crossing-point anchoring on PIE. Re-anchoring removes all window leakage and grows the sample 3.5-fold. Window counts are total (train/validation/test); the class weight is the training loss's positive-class weight, the training split's negative-to-positive ratio.

Protocol	Windows	Positive Rate	Window Leakage	Class Weight
Track-end anchor	1389 (616/186/587)	41.0%	387/570 (67.9%)	1.44
Crossing-point	4906 (2178/634/2094)	33.6%	0/4906 (0.0%)	1.682

3.4. Features and Preprocessing

Every model receives the same input: a 16-frame window (0.5 s at 30 fps) of a five-dimensional vector $[x_1, y_1, x_2, y_2, v]$, where (x_1, y_1, x_2, y_2) are the pedestrian's bounding-

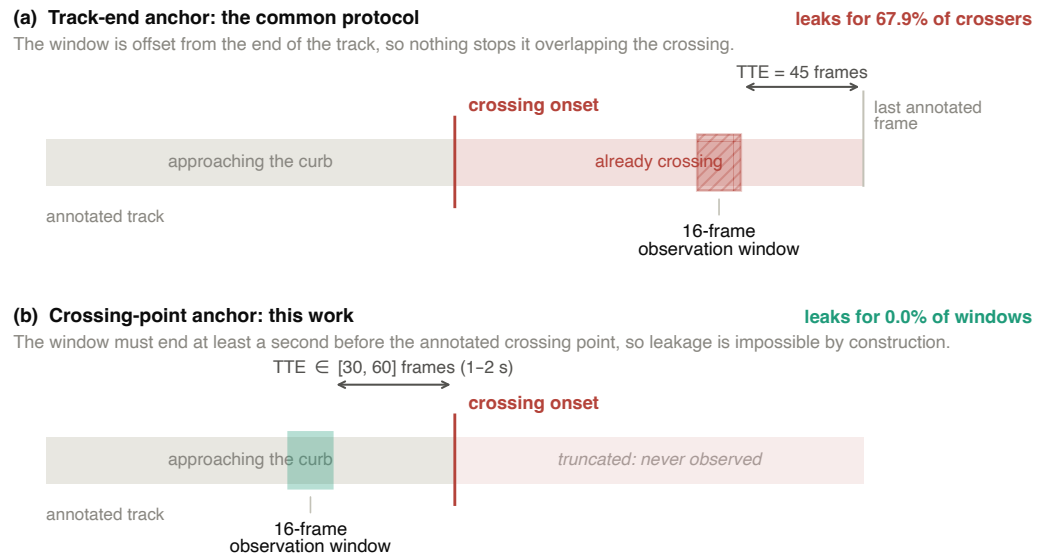


Figure 3. Where the observation window lands under each anchoring rule. (a) A track-end anchor places no constraint on the onset, so the window may overlap the crossing. (b) Anchoring at crossing_point with a 30–60 frame look-ahead makes leakage impossible.

box corners and v is the ego-vehicle speed. Box coordinates are kept in raw PIE pixel units on the 1920×1080 image plane and deliberately *not* normalized by image size, so the network sees absolute geometry. Each feature is standardized by its own z-score with μ and σ computed on the training split alone, since computing them over the whole dataset would leak distributional information about the evaluation sets. The ablation of Section 5.5 drops v and is otherwise identical. Unless stated otherwise the decision threshold is 0.5 on the sigmoid output, which makes our numbers directly comparable with the baselines in Table 4.

3.5. Model Architectures

All four families share one minimal wrapper: a linear projection of the five input features to 64 dimensions with a ReLU nonlinearity, a temporal encoder, a read-out of the final or pooled time step, and a linear head producing a single logit. Only the encoder differs, and every network is trained from scratch.

All three recurrent families are two-layer bidirectional networks with last-step read-out, and we refer to them by their cell: the **LSTM** [12] is the reference system, the **GRU** is that network with its cell replaced by a gated recurrent unit [29] and nothing else changed, isolating *which* gated cell is used, and the **RNN** is the same network with gating removed entirely, a bidirectional tanh (Elman) cell [35], isolating *gating itself*. They form a ladder: if recurrent machinery is what makes this task work, performance should fall at each rung. The **Transformer** is a small pre-layer-norm encoder [28] whose width, depth, head count, pooling, and positional encoding were searched. Two further LSTM variants serve as ablations, a bounding-box-only model dropping the ego-speed stream and a model with additive temporal attention over the recurrent states [36].

3.6. Metric Hierarchy, Evaluation, and Statistical Procedure

We report F1 first, accuracy second, and ROC-AUC and PR-AUC as threshold-free corroboration. The ordering suits an imbalanced, safety-critical binary decision: a deployed system must commit to a threshold, so a metric summarizing performance at an operating point is more informative than one averaging over all of them, and with roughly one window in three positive, accuracy alone rewards a model leaning toward the majority

class. AUC still checks whether an F1 result is an artifact of threshold placement. Making F1 primary has three consequences, applied symmetrically to all four families: τ is tuned on validation data only, each configuration is re-selected by validation F1, and checkpoints are taken at best validation F1. Models selected under the older AUC-first rule are labelled as such.

Point estimates are per-seed means over five seeds, and every claim of a difference is backed by an interval from a 10,000-resample bootstrap on the 2,094 test windows. Windows from the same pedestrian are not independent, since they overlap in time and share an identity, a scene, and a behaviour, so a bootstrap that resamples windows understates uncertainty; we therefore also report a *pedestrian-cluster* bootstrap that resamples the 541 test pedestrians as whole units and treat that stricter interval as the primary evidence. Comparisons between models are *paired* bootstraps on identical resamples, which removes the shared difficulty of the test set from the contrast. Generalization beyond the fixed split is assessed by leave-one-set-out (LOSO) cross-validation across all six recording sets. A five-seed probability ensemble, where reported, is labelled as such.

3.7. Live Perception-to-Prediction Pipeline

For the deployment analysis we built the pipeline of Figure 2b. Pedestrians are detected with a YOLO-family single-stage detector [31], associated across frames by ByteTrack [32], and buffered per track into the same 16-frame window of $[x_1, y_1, x_2, y_2, v]$ the offline models consume, with v read from the vehicle rather than from vision. It supplies the latency measurement (Section 5.7), the detector-in-the-loop experiment (Section 5.8), and the qualitative examples (Section 5.9); the benchmark results use annotated boxes, as every comparable study does.

4. Experimental Settings

Model training under the clean protocol, all statistical analysis, the recurrent-family searches, and every timing measurement ran locally on an Apple M4 system-on-chip (10 CPU cores, 16 GB unified memory, macOS 26.5). The larger Transformer search and the observation-length sweeps ran on a Kaggle instance with an NVIDIA Tesla T4 GPU (16 GB). Every configuration in a results table was retrained locally through the same engine, so no reported number depends on the machine that produced it; inference parity was verified at approximately 10^{-6} in output probability. The software stack was Python 3.13.5 with PyTorch 2.12.0 [37], NumPy 2.4.6, SciPy 1.17.1, and scikit-learn 1.9.0; the perception front end used Ultralytics 8.4.68 with a YOLO26-M detector and the bundled ByteTrack implementation.

One reproducibility caveat: training `nn.LSTM` on Apple's Metal Performance Shaders backend proved dependent on process history, so the same configuration and seed can give different results depending on what ran earlier in the process. Recurrent runs needing exact reproduction were therefore executed on CPU, where training is bit-reproducible; Transformer training did not show this behaviour.

Every family trains through a single model-agnostic engine, so the recipe cannot drift between architectures. The loss is binary cross-entropy with a positive-class weight of 1.682, the training split's negative-to-positive ratio, applied only in the training gradient. Optimization uses Adam [38] with weight decay 10^{-5} (decoupled in the Transformer recipe stage), batch size 32, at most 100 epochs, early stopping with patience 15, and a plateau schedule halving the learning rate after five epochs without validation improvement. Every configuration is trained with five seeds (42, 0, 1, 2, 3), and the test set is evaluated once per model, on the validation-selected checkpoint, by the one script with test access.

Hyperparameters are searched rather than asserted. The LSTM received a 36-configuration grid, fixed a priori, over learning rate $\{10^{-3}, 5 \cdot 10^{-4}, 10^{-4}\}$, dropout $\{0.2, 0.3, 0.5\}$, hidden size $\{64, 128, 256\}$, and depth $\{1, 2\}$, the 54 nominal cells collapsing to 36 because inter-layer dropout is inert at one layer. Selection ran in three stages: all 36 at seed 42, then the top five plus the baseline across five seeds, then the winner and baseline with the test set touched once. The GRU and the RNN each received the *identical* budget. The Transformer, having more architectural freedom, received a 78-configuration staged search over width $\{(64, 128), (128, 256), (128, 512)\}$, depth $\{2, 4\}$, pooling, and positional encoding, then a recipe stage over learning rate, schedule, dropout, and weight decay. No stage evaluated the test set. Table 2 lists what the searches selected; the three recurrent searches independently chose the same width and depth.

Because the searches select different settings for different families, the comparison is run twice. In the *tuned* design each family uses its own searched configuration under an equal budget. This follows the large-scale architecture audits, which tune every candidate separately and find that apparent architectural progress often reflects unequal tuning [39,41]; Melis et al. [40] show a plain LSTM overtaking newer architectures once every model receives the same budget. Imposing one family’s settings on another relocates that bias rather than removing it, since the shared setting suits whichever family produced it: our RNN’s search chose a learning rate an order of magnitude below the LSTM’s, and forcing the LSTM’s rate on the un-gated cell would handicap it and manufacture an advantage for gating. The *matched* design therefore runs the three recurrent families at one identical recipe, hidden size 128, two layers, dropout 0.3, learning rate 10^{-3} , AUC-selected checkpoint, with *only* the cell class changed; Section 5.3 reports both. Width and parameter count cannot both be matched, since at hidden size 128 an LSTM carries four gate matrices where an un-gated cell carries one: equal width means 594,561 parameters against 149,121. We match width, as a cell swap requires, whereas Chung et al. [42] match parameter count; parameters are reported throughout so either reading is available.

Table 2. The configuration each family’s search selected. Everything not listed is identical across families (batch 32, ≤ 100 epochs, patience 15, train-only z-score, Adam, weight decay 10^{-5} , class weight 1.682). “Width” is the recurrent hidden size, read-out being twice this, or the Transformer’s model and feed-forward dimensions; τ is tuned on validation only.

Model	Cell	Width	Layers	Drop.	LR	τ
LSTM	LSTM	256	2	0.3	10^{-3}	0.52
Transformer	attention (4 heads)	128/512	4	0.1	10^{-3}	0.65
GRU	GRU	256	2	0.3	$5 \cdot 10^{-4}$	0.53
RNN	Elman (tanh)	256	2	0.2	10^{-4}	0.53

5. Results

5.1. The Temporal-Leakage Audit

Auditing the track-end protocol against the recovered per-frame cross state gives the paper’s first finding. Of 570 crossing pedestrians, 387 (67.9%) have at least one frame inside the observation window in which they are already crossing and 369 (64.7%) are crossing in *all sixteen*; only 183 (32.1%) have a clean window (Figure 4a). The median gap between the end of the window and the crossing onset is +182 frames, six seconds *into* the crossing. For roughly two-thirds of the positive class, what is reported as prediction is the detection of an event already under way.

A static shortcut accompanies the leakage. The single bounding box at the last observed frame separates the classes on its own, with no temporal information at all: rank-

biserial correlations of +0.65 for box area, +0.63 for height, and +0.49 for the bottom edge (Figure 4b), as expected of pedestrians already in the road and therefore nearer the camera. Models trained on this protocol converged by epoch 3. The identical audit on the crossing-point protocol finds 0 of 4,906 windows leaking and every static effect below 0.3 (area 0.25, height 0.21, bottom edge 0.09), so whatever signal remains must live in temporal dynamics.

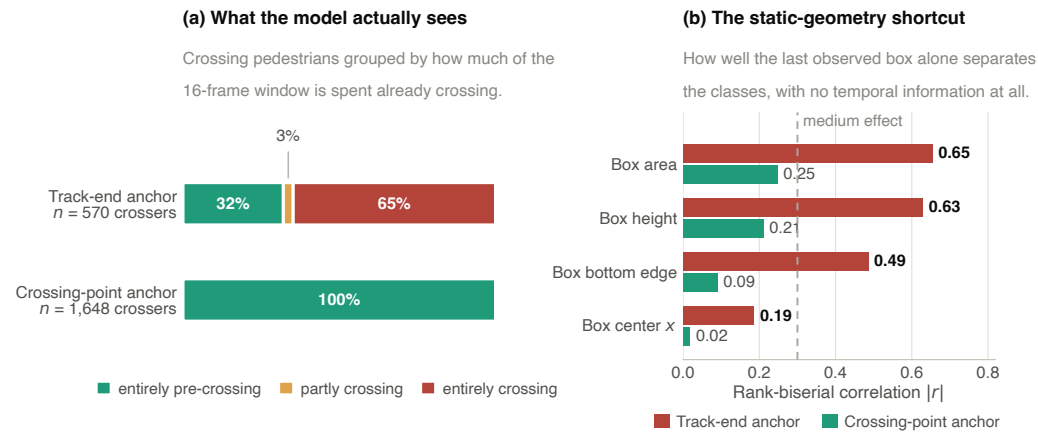


Figure 4. The temporal-leakage audit, from PIE’s own per-frame cross annotations. (a) Crossers grouped by how much of the window is already spent crossing. (b) How well the last observed box alone separates the classes.

5.2. Main Result

On the leakage-free protocol the two-stream LSTM reaches test F1 0.844, accuracy 0.897, and ROC-AUC 0.940 under F1-first optimization (Table 3), with PR-AUC 0.876 and a 95% bootstrap AUC interval of $[0.92, 0.95]$ at the window level and $[0.92, 0.96]$ over pedestrian clusters.

Removing the leakage barely moved this model’s AUC, from 0.931 on the leaky protocol to 0.932 ± 0.011 over five seeds, while moving convergence from epoch 3 to a believable epoch 17; the correction is methodological rather than a deflation of the score. Nor is the new protocol simply an easier one: per-pedestrian AUC (0.914) matches per-window AUC (0.913), so the result is not an artifact of correlated windows, and the short tracks admitted by the 50% overlap are *harder* than the rest (AUC 0.863 for tracks of 46–75 frames against 0.919 for longer ones).

5.3. Does the Architecture Matter?

Table 3 reports every family under both designs of Section 4, and Figure 5 gives the paired contrasts with pedestrian-cluster intervals. The two designs agree. Held at the matched configuration, where only the recurrent cell class changes, F1 is 0.828 for the LSTM, 0.840 for the GRU, and 0.836 for the RNN; the un-gated cell reaches that with 149,121 parameters against the LSTM’s 594,561, carrying one weight matrix where an LSTM carries four. Tuned per family at an equal budget, F1 is 0.844, 0.847, 0.849, and 0.852 for the LSTM, Transformer, GRU, and RNN. Tuning lifts every family by a comparable margin and reorders none of them.

On ROC-AUC the Transformer does separate. Selecting its checkpoint on validation AUC rather than validation F1 gives 0.950, the best figure obtained, and a paired bootstrap on identical test windows confirms that this AUC-selected Transformer beats the AUC-selected LSTM ($\Delta\text{AUC} = +0.0135$, 95% CI $[+0.0097, +0.0174]$). The margin belongs to the

search rather than to attention: the same family run at the matched configuration instead of its searched one merely ties the LSTM ($\Delta\text{AUC} = +0.0005$, interval spanning zero).

On the primary metric, however, **the four families are statistically indistinguishable**. Every cross-family ΔF1 interval spans zero (Figure 5a): GRU against LSTM (+0.0071), RNN against LSTM (+0.0033), Transformer against LSTM (+0.0008), and each remaining pair likewise. Removing the LSTM's gating entirely costs nothing over a 0.5 s window; the un-gated network even ties the AUC-selected Transformer on AUC ($\Delta\text{AUC} -0.0013$, interval spanning zero) while being the smallest and fastest family in the study.

These ties are not a test without power: the same procedure detects differences where they exist, each F1-optimized model beating its own family's AUC-selected baseline by an interval clearing zero (Figure 5a, lower group), and several AUC contrasts separating cleanly (Figure 5b).

Table 3. Every family under both comparison designs, on the leakage-free PIE test set (set03; 2,094 windows, 32.5% positive). Values are per-seed means over five seeds. The two designs agree: no cross-family difference in either block is significant on F1 (Figure 5a). (†) attention has no hidden-size equivalent, so the matched Transformer is the same recipe adapted to a two-layer, 128/256 encoder, and is the one row selected on validation F1 rather than validation AUC. Selecting the tuned Transformer on validation AUC instead raises its AUC to 0.950, the best figure obtained.

Model	Params	F1	Acc	AUC
<i>Matched configuration: width 128, 2 layers, dropout 0.3, LR 10^{-3}; only the cell changes</i>				
LSTM	594,561	0.828	0.883	0.932
GRU	446,081	0.840	0.898	0.933
RNN	149,121	0.836	0.889	0.942
Transformer [†]	268,417	0.821	0.878	0.942
<i>Tuned per family at an equal search budget (Table 2)</i>				
LSTM	2,237,313	0.844	0.897	0.940
Transformer	794,241	0.847	0.896	0.947
GRU	1,678,209	0.849	0.901	0.941
RNN	560,001	0.852	0.902	0.948
<i>Input and read-out ablations (LSTM, matched configuration)</i>				
LSTM, bounding box only (4-D)	594,497	0.551	0.744	0.753
LSTM + temporal attention	611,265	0.821	0.879	0.925

5.4. Comparison with Published Baselines

Table 4 places our models beside published PIE results on the standard protocol, with a column counting input streams. On F1 the two-stream models reach 0.844–0.852, ahead of most multimodal baselines and within about 0.02 of the ceiling set by PedFormer (0.87), using two inputs instead of three to seven; on AUC they lead the table at 0.94–0.95; on accuracy they sit mid-band, 0.90 against PedFormer's 0.93. PIP-Net [6] and Liu et al. [21] are discussed but not tabulated, the first using a custom split and the second an occlusion-specific, roughly one-frame-ahead protocol.

5.5. Which Input Carries the Signal?

Dropping the ego-speed stream from an otherwise identical LSTM collapses it: AUC falls from 0.932 to 0.753, F1 from 0.828 to 0.551, and accuracy from 0.883 to 0.744. A loss of 0.18 AUC from removing one of two inputs identifies ego-vehicle speed as the dominant predictor, consistent with the independent feature-importance analysis of Azarmi et al. [8]; adding temporal attention to the same model, by contrast, changes nothing (AUC 0.925 against 0.932). The collapse is also the other half of the leakage story, since once the static-

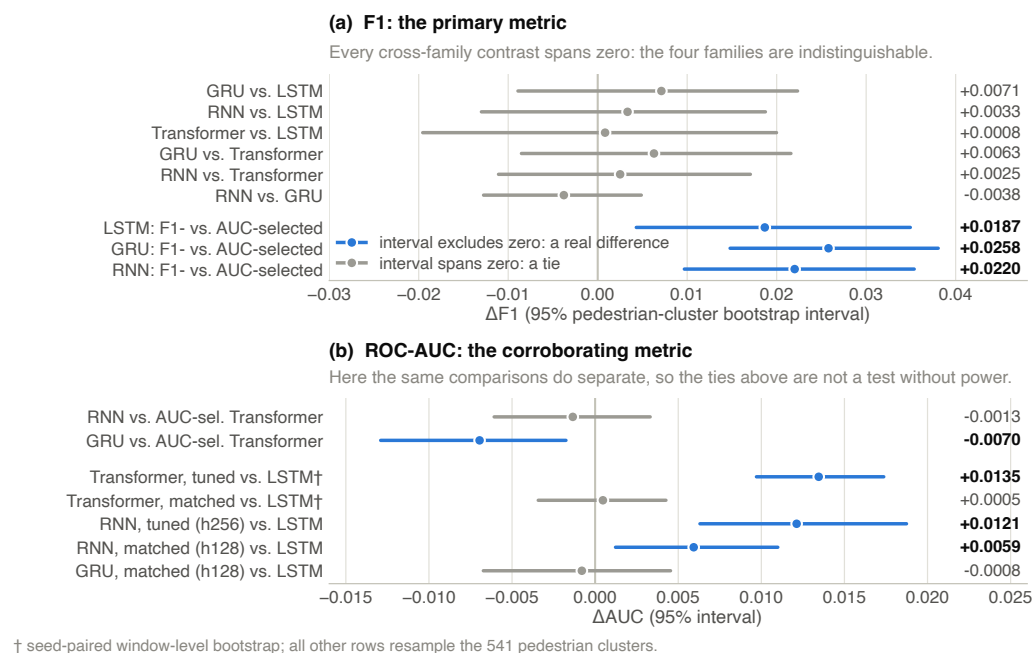


Figure 5. Paired contrasts on identical test windows, with 95% pedestrian-cluster bootstrap intervals. (a) On F1 every cross-family contrast spans zero; the lower group shows the same test detecting a real effect. (b) On ROC-AUC the models separate, with matched-configuration contrasts shown beside tuned ones.

Table 4. Crossing-prediction performance on PIE under the standard protocol (train set01/02/04, validate set05/06, test set03). Baseline figures are as reported by the cited sources. Our rows are per-seed means over five seeds at threshold 0.5.

Method	Acc	AUC	F1	Modalities (Streams)
PCPA [4]	0.87	0.86	0.77	box + pose + context + speed (4)
Pedestrian Graph+ [14]	0.89	0.90	0.81	pose graph + ego (2–3)
IntFormer [17]	0.89	0.92	0.81	multimodal
PIT [18]	0.91	0.92	0.82	multimodal
BiPed [19]	0.91	0.90	0.85	multimodal
PedFormer [19]	0.93	0.90	0.87	multimodal (multitask)
GTransPDM [7]	0.90	0.87	0.82	box + pose + ego (3)
GTransPDM without pose [7]	0.92	0.90	0.86	box + ego (2)
PedCMT [20]	0.92	—	—	box + ego-speed (2)
MFT [24]	0.90	0.94	0.83	4 context streams
LSTM (ours)	0.897	0.940	0.844	box + ego-speed (2)
Transformer (ours)	0.896	0.947	0.847	box + ego-speed (2)
GRU (ours)	0.901	0.941	0.849	box + ego-speed (2)
RNN (ours)	0.902	0.948	0.852	box + ego-speed (2)

geometry shortcut is removed a model given only bounding boxes has very little left to work with.

5.6. Prediction Horizon and Observation Length

How far ahead the model must predict matters; how long it gets to observe does not. AUC declines monotonically as the horizon lengthens, from 0.961 at 1.0 s to 0.946 at 1.5 s and 0.919 at 2.0 s, with every pairwise paired t -test significant ($p \leq 0.004$) and Kruskal–Wallis $p = 0.002$. Because a longer horizon needs a longer usable track and so changes

which pedestrians are eligible, we repeated the comparison on a matched cohort of the same 493 test pedestrians; the decline survives unchanged (sample effect ≤ 0.002 AUC). This is what one wants from a genuine predictor, and it corrects an earlier “insensitive to horizon” conclusion of our own that was itself a leakage artifact.

Observation length behaves in the opposite way. Over 8, 16, and 30 frames the baseline’s AUC moves only from 0.931 to 0.933 to 0.937, a spread smaller than seed noise (all pairwise $p > 0.21$), and extending the window to 32 and 64 frames makes matters slightly worse for every family. Half a second already carries the available dynamics, which justifies the 16-frame window and suits deployment, since a shorter window means a shorter wait before a newly detected pedestrian can be scored. One asymmetry is worth flagging: at 64 frames the un-gated RNN alone loses 0.050 F1 against its own 16-frame result, roughly double the decline of the gated cells. Per-seed intervals overlap, so this is directional rather than bootstrap-confirmed, but the equivalence we report is horizon-bounded.

5.7. Cross-Set Generalization, Capacity, and Latency

Rotating the held-out recording set across all six folds gives an AUC of 0.928 ± 0.041 with no fold collapsing. The set03 fold, our fixed test set, scores 0.931, at the fold mean rather than above it, so the headline number does not depend on set03 being an unusually easy partition; excluding only the 47-window set05 fold, too small to interpret, the remaining five give 0.915 ± 0.029 .

A hidden-size sweep (64, 128, 256 units giving AUC 0.927, 0.933, 0.938) and a depth sweep (1, 2, 3 layers giving 0.930, 0.932, 0.931) show the recurrent baseline is limited by data rather than capacity: hidden size 256 is not significantly better than 128 ($p = 0.34$) despite 3.8 times the parameters. The 36-configuration grid agreed, its validation winner beating the hand-set configuration on test by only $+0.0006$ ($p = 0.91$).

Latency does not discriminate among the families. On the Apple M4 CPU at batch 1 the isolated models run in 0.316–0.721 ms per window, between 46 and 105 times inside a 30 fps frame budget, the un-gated RNN fastest. In the full pipeline the detector accounts for about 93% of per-frame cost and the intention model for 4.5%, so the frame rate is a property of the detector, not of the crossing-intention model.

5.8. Detector-in-the-Loop Robustness

Benchmark numbers on PIE are computed from annotated boxes; a deployed system receives boxes from a detector. On two test-set clips covering 98 pedestrians and 311 windows, replacing annotated boxes with detector boxes (mean IoU 0.75) moved AUC only from 0.962 to 0.953 per window and from 0.958 to 0.948 per pedestrian, with the decision flipping in 10 of 311 windows (3%): prediction is robust to realistic box noise. The weak links lie earlier. Detector recall is 88% of pedestrians, so roughly one in eight is never covered well enough to be scored, and identity fragmentation is worse, a pedestrian’s dominant ByteTrack identity covering a mean of only 39% of its matched frames. One caveat bounds the experiment: each detector window uses the best-IoU detection per frame matched against the annotated box, which isolates localization noise but removes the association errors just quantified, so the 0.009 AUC drop is a lower bound on end-to-end degradation rather than an estimate of it.

5.9. Qualitative Behaviour

Figure 6 shows the complete pipeline on two test-set scenes, with boxes from the detector and tracker rather than from annotations and probabilities from the five-seed LSTM ensemble at its validation-tuned threshold of 0.516. Panel (a) is the leakage-free protocol made visible: the model commits to “will cross” at $p = 0.71$ a second and a half before this pedestrian steps off the kerb. Panel (b) is the complementary and harder

case; to any rule based on position it is the same picture, a person standing at a kerb beside a marked crosswalk, but this one does not cross and the model correctly leaves him at $p = 0.31$. What separates the panels is the motion history behind each box, not where either pedestrian stands. Both are annotated in PIE, so each verdict is checked against ground truth rather than judged by eye; the frames are illustrative rather than representative, and across the two clips the ensemble scores accuracy 0.920 and F1 0.884 over 439 windows. Video S1 shows the pipeline running continuously over 26 s of test video.

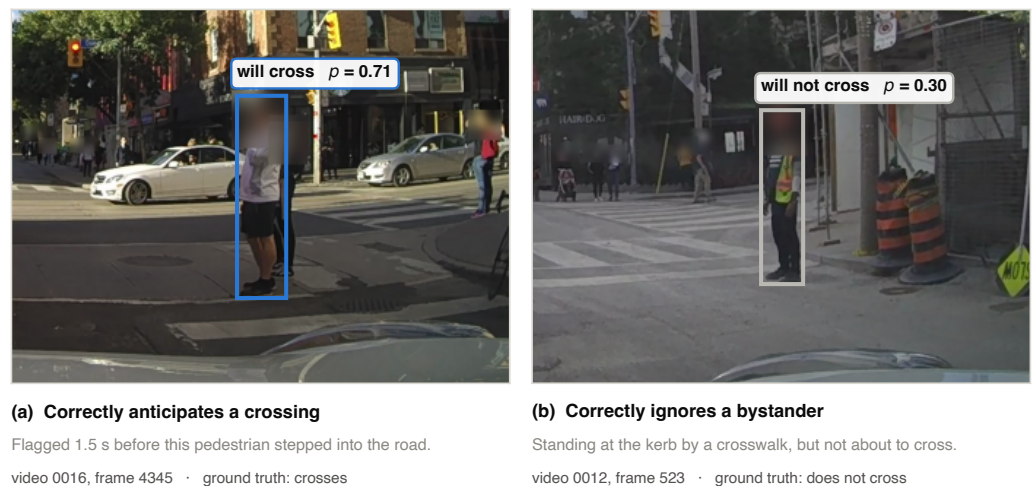


Figure 6. The detection-to-prediction pipeline on two PIE test scenes, with boxes from YOLO and ByteTrack. (a) A pedestrian correctly flagged 1.5 s before stepping into the road. (b) A worker at a kerb, correctly not flagged. Faces were blurred before publication.

6. Discussion

Two cheap inputs reach F1 within about 0.02 of the multimodal ceiling and the highest AUC in the standard-protocol table, at a small fraction of the feature-extraction cost. We do not claim to be first to find this: PedCMT [20] and the pose-free ablation of GTransPDM [7] reported comparable results from the same two streams, and Achaji et al. [30] made the case from box geometry alone. What this paper adds is a demonstration that the finding survives a leakage-free protocol and full statistical scrutiny, together with an account of *why* it holds. Removing ego-speed collapses the model, while replacing its temporal encoder with three quite different alternatives changes nothing measurable; the predictive content is therefore concentrated in the ego-speed profile and coarse box dynamics, legible within half a second, leaving no long-range structure for a heavier encoder to find. That is a claim about the task rather than about our implementation, and it explains why so many differently architected models cluster within a few points of one another. For a practitioner the implication is direct: effort spent on a pose estimator or an optical-flow network is likely better spent on the perception front end.

The architecture comparison resolves differently depending on which metric leads: on AUC the Transformer wins, on F1 all four families tie. Neither result is wrong, and a paper reporting only one would be telling half the story. Given the F1-first hierarchy and near-identical latency there is no performance argument for preferring a Transformer here, and several for preferring a small recurrent model. The methodological consequence is sharper: an architecture comparison reporting a single metric, from a single seed, with unequal search budgets can produce a headline in either direction. Ours could have.

That two-thirds of crossing pedestrians were observed mid-crossing under a widely used windowing rule is the most transferable finding here, because nothing about it is specific to our model: any study anchoring its window at the end of a track inherits it. Removing the leakage barely changed the headline AUC, which is the point rather than a reassurance, since a model scoring 0.93 by recognizing crossings in progress and one scoring 0.93 by anticipating them a second in advance are different systems that happen to share a digit. Three symptoms may help others recognize the artifact: implausibly fast convergence, epoch 3 rather than 17; static box geometry separating the classes on its own at effect sizes above 0.6; and insensitivity to the prediction horizon, the clearest tell, since a model that genuinely respects a horizon must find a longer one harder. The audit is cheap, needing only the per-frame event annotation the dataset already provides.

Several limitations bound these conclusions. The first is that they rest on a single dataset: PIE is the field's standard benchmark, but a protocol artifact demonstrated on one corpus is a hypothesis about the others until it is checked, and the same audit on a second dataset is the most direct extension of this work. The dominant feature is also partly a measurement of the driver, since an instrumented vehicle slows for a pedestrian the driver expects to cross; ego-speed is legitimate at inference time and available on any production vehicle, but the field has independently flagged this driver-side bias in yielding scenarios [8], and a speed-perturbation study, which we have not run, is the natural next step. The equivalence is horizon-bounded, since the un-gated network begins to fall behind at a two-second window, so "gating is unnecessary" is a statement about half a second, not about recurrent architectures in general. The detector-in-the-loop study rests on two clips and an oracle-guided association that makes its degradation a lower bound, and seed-level statistical power is low, which is why the paired and pedestrian-cluster bootstraps carry the argument.

7. Conclusions

We revisited pedestrian crossing-intention prediction on PIE and found that a widely used windowing protocol lets the crossing event into the observation window for 67.9% of crossing pedestrians, turning much of the benchmark into detection rather than prediction. Re-anchoring at PIE's own crossing point removes the leakage entirely and enlarges the dataset 3.5-fold.

On that protocol a model using only a pedestrian bounding box and the ego-vehicle's speed reaches F1 0.844–0.852 and the highest AUC among published results on the standard split, with two input streams rather than the three to seven used by recent multimodal systems. Ego-speed is the dominant predictor, since removing it costs 0.18 AUC. The temporal architecture, by contrast, is close to irrelevant: an LSTM, a Transformer, a GRU, and an un-gated RNN, trained under one identical engine, are statistically indistinguishable on the primary metric, and even removing gating costs nothing over a half-second window. The input signal, not the architecture, decides. No comparison rests on a point estimate; each is backed by a paired bootstrap over pedestrian clusters, with the headline model further checked by leave-one-set-out cross-validation, five-seed variance, a documented search, and a detector-in-the-loop test.

Three directions follow. The first is cross-dataset work: repeating the leakage audit on other crossing benchmarks would show whether the artifact belongs to the windowing convention rather than to PIE, and validating the full two-stream model elsewhere requires a dataset carrying synchronized ego-motion. The second is a speed-perturbation study, probing how far the ego-speed result depends on the driver's own anticipation. The third is the perception front end, since detector recall and track identity, not the intention model, would limit such a system in a vehicle.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/mti1010000/s1>: Video S1, the detection-to-prediction pipeline running on PIE test clip video_0012 (26 s, annotated with per-track crossing probabilities and ego-vehicle speed).

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Institutional Review Board Statement: Not applicable. This study used the publicly available PIE dataset [3] and involved no new experiments on humans or animals. The face region of every person detected in the PIE frames reproduced in Figure 6 was irreversibly blurred before publication.

Informed Consent Statement: Not applicable.

Data Availability Statement: The PIE dataset [3] is publicly available. The extraction protocol, training engine, and evaluation code developed here are available from the authors on reasonable request. (PLACEHOLDER: replace with the repository URL upon release.)

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